

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I SAHASRA V(192424234) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Sahasra V

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

Scenario: Smart Farming Crop Disease Detection System

A machine learning system is developed to detect diseases in crop leaves. Farmers upload images of crop leaves, and the system predicts whether the crop is healthy or affected by a disease.

Functional Blocks of a Machine Learning System

1. **Data Collection**
 1. Collect crop leaf images from farms and agricultural datasets.
 2. Data may be **labelled** (Healthy, Rust, Blight) or **unlabelled** (only images).
2. **Data Preprocessing**
 1. Remove noisy or blurred images.
 2. Resize images and normalize pixel values.
 3. Handle missing or duplicate data.
3. **Feature Extraction**
 1. Extract useful information such as leaf color, texture, shape, and spots.
 2. These features help the model identify diseases.
4. **Model Training**
 1. **Supervised Learning:** Uses labelled images to train a classifier (e.g., Healthy or Diseased).
 2. **Unsupervised Learning:** Uses unlabelled images to group similar disease patterns into clusters.
5. **Model Evaluation**
 1. Test the trained model using unseen data.
 2. Measure performance using **Accuracy, Precision, Recall, and F1-score**.

Working Example

Supervised Learning

- **Input:** Images labelled as Healthy, Rust, or Blight.
- **Model:** Decision Tree or CNN.
- **Output:** Predicts the disease category of a new leaf image.

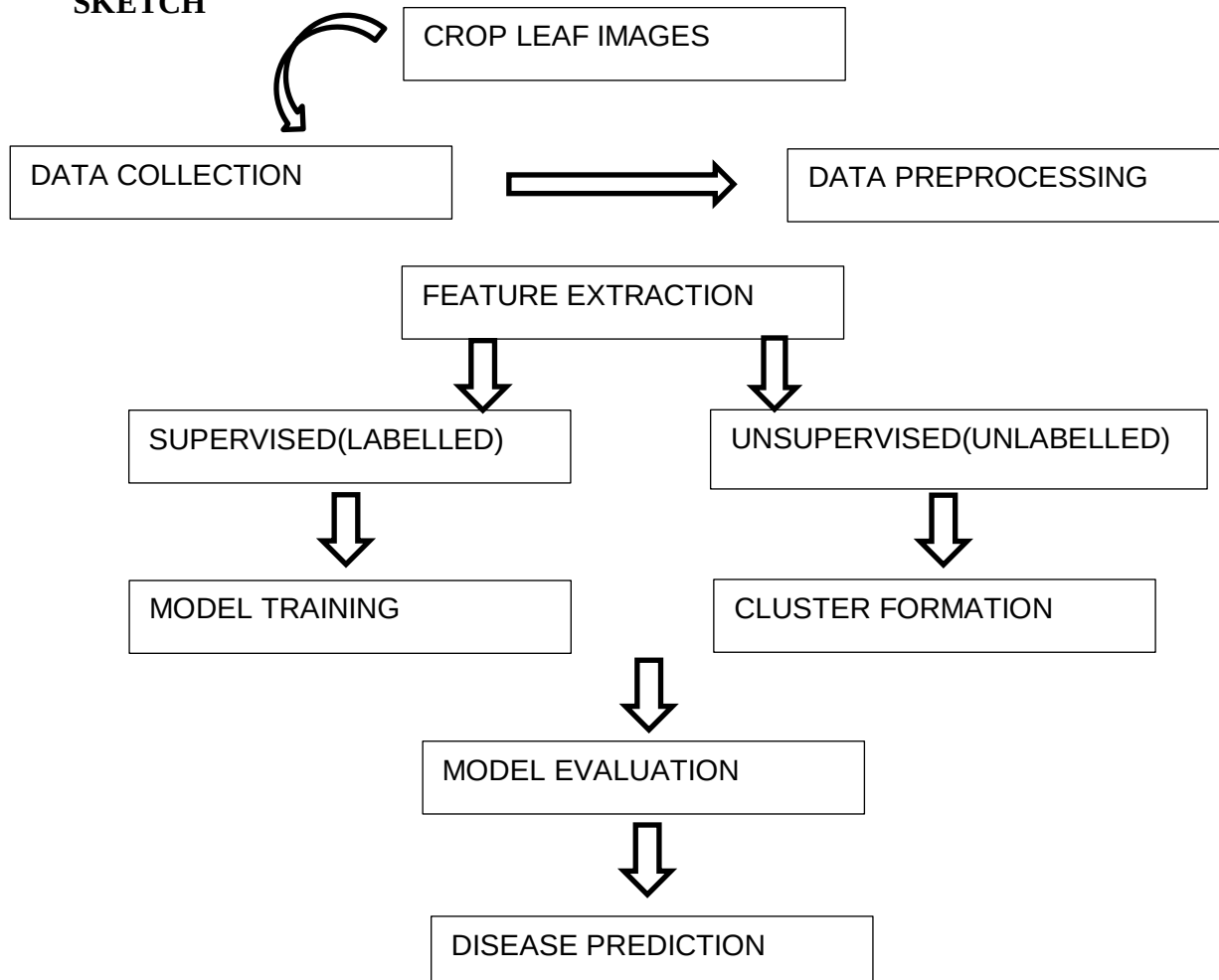
Handling of Data: Uses **labelled data** where the correct disease name is already known.

Unsupervised Learning

- **Input:** Crop leaf images without labels.
- **Model:** K-Means Clustering.
- **Output:** Groups similar leaves into clusters based on visual characteristics.

Handling of Data: Uses **unlabelled data** to discover hidden patterns without predefined classes.

SKETCH



2.Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any

other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

Scenario

A government disaster management agency wants to predict **floods and earthquakes** using historical records and real-time sensor data. The system provides early warnings to help authorities reduce damage and save lives.

1. Data Collection

The system collects data from multiple sources:

- Historical flood and earthquake records
- Rainfall and weather data
- River water level sensors
- Seismic sensor readings
- Satellite images
- GPS and IoT sensor data

Both **historical** and **real-time** data are continuously updated.

2. Data Preprocessing

Before training the model, the data is cleaned and prepared.

- Remove duplicate records
- Handle missing values
- Remove noisy sensor readings
- Normalize numerical values
- Convert categorical data into numerical format
- Synchronize data collected from different sources

These steps improve prediction accuracy.

3. Feature Selection

Important features are selected to improve model performance.

Flood Prediction Features

- Rainfall
- River water level
- Soil moisture
- Temperature
- Humidity

Earthquake Prediction Features

- Seismic wave magnitude
- Ground vibration
- Fault-line distance
- Previous earthquake history
- Depth of earthquake

Feature selection techniques:

- Correlation Analysis
- Information Gain
- Principal Component Analysis (PCA)

Selecting only important features reduces computation and increases accuracy.

4. Model Selection

Recommended Model: Neural Network (Deep Learning)

Justification

- Handles large and complex datasets.
- Learns non-linear relationships between environmental factors.
- Works well with real-time sensor data.
- Provides higher prediction accuracy for complex disasters.

Alternative Models

- Random Forest – Good for flood prediction with structured data.
- Decision Tree – Easy to understand but less accurate.
- Support Vector Machine (SVM) – Suitable for smaller datasets.

Why Neural Network?

It can learn complex patterns from weather, seismic, and satellite data better than traditional algorithms.

5. Training, Validation and Evaluation

Training

- Split the dataset into **80% training** and **20% testing**.
- Train the neural network using historical disaster data.

Validation

- Use a validation dataset to tune model parameters.
- Prevent overfitting by monitoring validation accuracy.

Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

If the model performs well, it is deployed for real-time prediction.

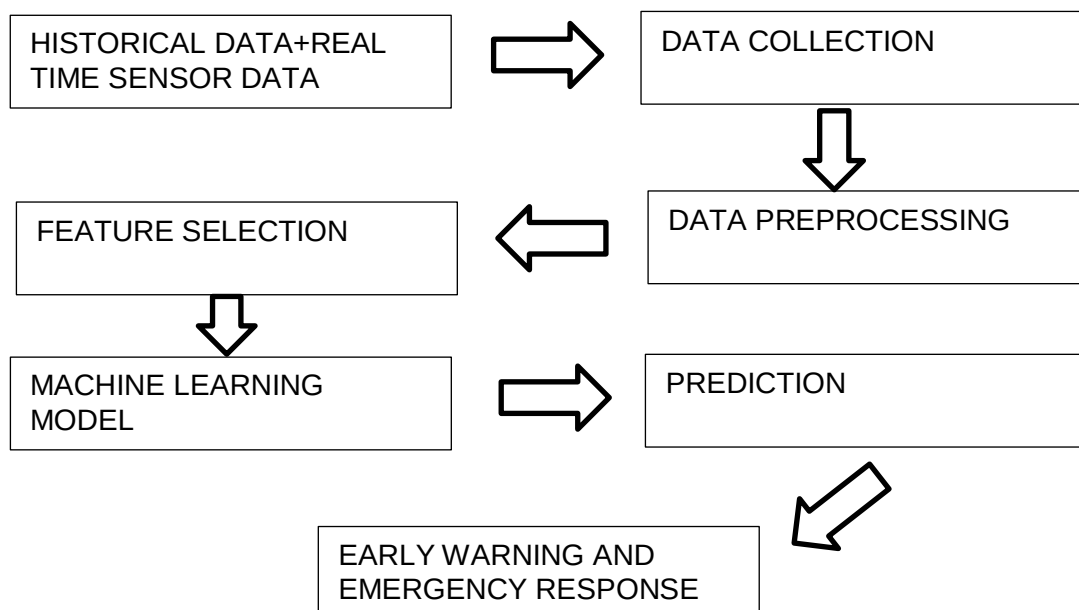
6. Working of the System

1. Collect real-time weather and seismic data.
2. Preprocess and clean the data.
3. Extract important features.
4. Feed the data into the trained neural network.
5. Predict flood or earthquake risk.
6. Send alerts to disaster management authorities.

7. Applications in Disaster Management

The system helps authorities by:

- Providing **early warning** before disasters occur.
- Identifying **high-risk areas**.
- Supporting **risk assessment**.
- Planning evacuation routes.
- Allocating rescue teams efficiently.
- Reducing loss of life and property.
- Improving emergency response and resource management.



3. In the context of neural networks, consider a multilayer perceptron used for binary

classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

Scenario

A bank wants to predict whether a loan application should be **Approved (1)** or **Rejected (0)** based on customer details such as income, credit score, and loan amount. A **Multilayer Perceptron (MLP)** is trained using the **Backpropagation algorithm** to improve prediction accuracy

1. Forward Pass

- Input data (Income and Credit Score) is given to the input layer.
- The hidden layer processes the inputs using weights and an activation function.
- The output layer predicts either:
 - **1** → **Loan Approved**
 - **0** → **Loan Rejected**

Example:

- Actual Output = **1**
- Predicted Output = **0.80**

2. Error Calculation

The prediction is compared with the actual value.

Error = Actual Output – Predicted Output

Example:

- Actual = **1**
- Predicted = **0.80**

Error = 1 – 0.80 = 0.20

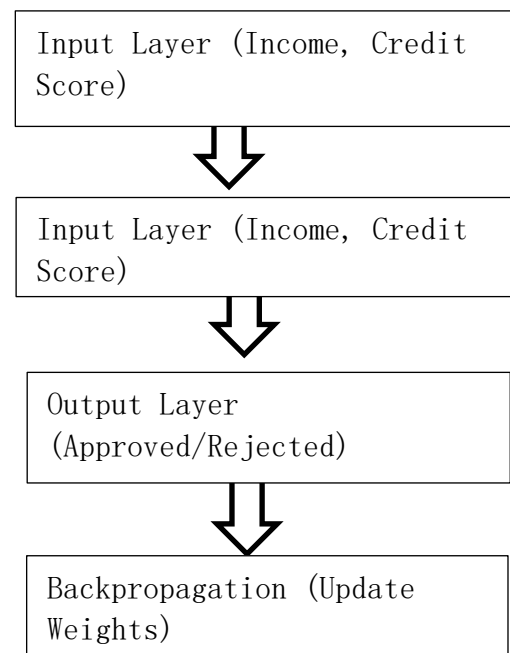
This error shows how far the prediction is from the correct output.

3. Backward Propagation

- The calculated error is propagated backward from the output layer to the hidden layer.
- The network computes how much each weight contributed to the error.
- The weights are adjusted to reduce future errors.

Weight Update Formula:

New Weight = Old Weight + Learning Rate × Error × Input



After updating the weights, the network performs another forward pass. This process repeats until the error becomes very small.

4. Role of Learning Rate

The **learning rate (η)** controls how much the weights are updated during training.

- **High Learning Rate**
 - Faster learning.
 - May overshoot the optimal solution.
 - Can cause unstable training.
- **Low Learning Rate**
 - More stable learning.
 - Slower convergence.
 - May require many training iterations.

A suitable learning rate helps the model converge faster with better accuracy.

5. Role of Activation Functions

Activation functions introduce non-linearity, allowing the network to learn complex patterns.

Common activation functions:

- **Sigmoid**
 - Output between 0 and 1.
 - Suitable for binary classification.
- **ReLU (Rectified Linear Unit)**
 - Faster training.
 - Reduces the vanishing gradient problem.
 - Commonly used in hidden layers.

Activation functions improve the model's learning capability and prediction accuracy.

Working Process

1. Input customer data into the network.
2. Perform the **forward pass** to obtain the predicted output.
3. Calculate the prediction error.
4. Propagate the error backward through the network.
5. Update the weights using the learning rate.
6. Repeat the process until the model achieves good accuracy.

4. In the context of Genetic Algorithms, consider a dataset representing different

candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution

Scenario

A company wants to develop an AI system that achieves **maximum Speed and Accuracy**. Each candidate solution is evaluated using a **fitness value**. A **Genetic Algorithm (GA)** is used to find the best solution through **selection, crossover, and mutation**.

Step 1: Initial Population

The initial population consists of the five candidate solutions.

Candidate	Fitness
1	40
2	60
3	75
4	65
5	80

Step 2: Selection

Selection chooses individuals with **higher fitness values** because they are more likely to produce better offspring.

Selected candidates:

- Example 5 → Fitness = **80**
- Example 3 → Fitness = **75**

These are the fittest individuals.

Step 3: Crossover

Crossover combines the characteristics of the selected parents.

Parent 1

- Speed = Medium
- Accuracy = High

Parent 2

- Speed = High
- Accuracy = Medium

After crossover:

Offspring	Speed	Accuracy
Child 1	High	High
Child 2	Medium	Medium

Step 4: Mutation

Mutation introduces a small random change to maintain diversity and avoid getting stuck in a local optimum.

Example:

Child 1

Before Mutation:

- Speed = High
- Accuracy = High

After Mutation:

- Slight parameter adjustment (e.g., tuning model settings), while retaining **High Speed** and **High Accuracy**.

Estimated Fitness:

Fitness = 90

Step 5: New Generation

Candidate	Speed	Accuracy	Fitness
Child 1	High	High	90
Child 2	Medium	Medium	60

The new generation contains a better candidate than the original population.

Best Optimal Solution

Speed	Accuracy	Fitness	Suitable
High	High	90	Yes

Role of Genetic Algorithm Operations

Selection

- Chooses individuals with the highest fitness values.
- Ensures better candidates are more likely to reproduce.

Crossover

- Combines features from two parent solutions.
- Produces offspring with improved characteristics.

Mutation

- Introduces small random changes.
- Increases diversity and helps explore the search space.
- Prevents the algorithm from getting stuck in local optima.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks) Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model’s accuracy and reliability. (5 Marks)

Scenario

A hospital wants to predict whether a patient has a disease based on **Age, Blood Pressure, Glucose Level, and Lifestyle Risk**. A **Neural Network (Multilayer Perceptron)** is trained using **Backpropagation** to classify patients as **Disease Present (Yes)** or **Disease Absent (No)**.

1. Confusion Matrix Calculation

Example	Actual	Predicted	Result
1	Yes	Yes	True Positive (TP)
2	Yes	No	False Negative (FN)
3	No	No	True Negative (TN)
4	Yes	Yes	True Positive (TP)
5	No	Yes	False Positive (FP)

Values

- True Positive (TP) = 2
- True Negative (TN) = 1
- False Positive (FP) = 1
- False Negative (FN) = 1

3. Evaluation Metrics

Accuracy

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ &= \frac{2 + 1}{2 + 1 + 1 + 1} = \frac{3}{5} = 0.60\end{aligned}$$

Accuracy = 60%

Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2}{2 + 1} = \frac{2}{3} = 0.667$$

Precision = 66.7%

Recall

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2}{2 + 1} = \frac{2}{3} = 0.667$$

Recall = 66.7%

F1-Score

$$\begin{aligned}\text{F1} &= \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \\ &= \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} = 0.667\end{aligned}$$

F1-Score = 66.7%

4. Role of Preprocessing

Preprocessing improves data quality before training.

- Remove missing values.
- Remove duplicate records.
- Normalize Age, Blood Pressure, and Glucose values.
- Encode categorical values (High, Medium, Low) into numerical form.
- Remove noisy or inconsistent data.

These steps improve the learning process and prediction accuracy.

5. Role of Feature Selection

Feature selection identifies the most important attributes affecting disease prediction.

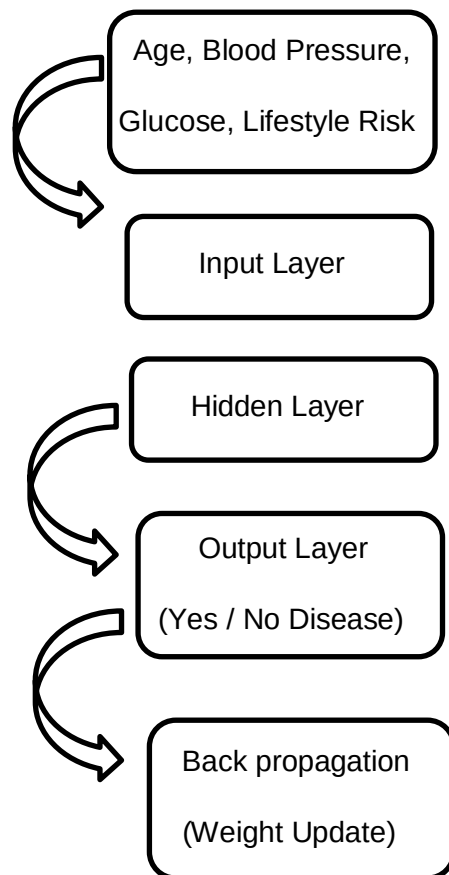
Important features include:

- Blood Pressure
- Glucose Level
- Lifestyle Risk
- Age

Benefits:

- Reduces unnecessary features.
- Improves model accuracy.
- Reduces training time.
- Prevents overfitting.

SKETCH DIAGRAM



6. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

1. Data Collection

The system collects traffic-related data from multiple sources:

- CCTV cameras
- Traffic sensors
- GPS from vehicles
- Weather stations
- Road network database

Input Features:

- Vehicle Density
- Time of Day
- Weather Conditions
- Road Type

2. Choice of Algorithm

Recommended Algorithm: Random Forest

Justification:

- Handles both numerical and categorical data.
- Gives high accuracy for traffic prediction.
- Less prone to overfitting than a single Decision Tree.
- Works well with large datasets and multiple input features.

Alternative Algorithms:

- Decision Tree – Simple but less accurate.
- Neural Network – Suitable for very large and complex traffic datasets.
- Support Vector Machine (SVM) – Better for smaller datasets.

Why Random Forest?

It provides accurate predictions, is easy to train, and performs well for real-time traffic classification.

3. Data Preprocessing

Before training, the collected data is cleaned.

- Remove duplicate records.
- Handle missing values.
- Remove noisy sensor readings.
- Normalize numerical values (vehicle density).
- Encode categorical variables (weather and road type).

These steps improve prediction accuracy.

4. Feature Engineering

Feature engineering creates useful inputs for the model.

Examples:

- Peak Hour (Yes/No)
- Average Vehicle Speed
- Traffic Density Index
- Rainfall Level
- Road Capacity Utilization

Benefits

- Improves prediction accuracy.
- Reduces unnecessary information.
- Makes the model more efficient.

5. Training the Model

- Split the dataset:
 - **80% Training**
 - **20% Testing**
- Train the Random Forest model using historical traffic data.
- Learn the relationship between traffic conditions and congestion levels.

6. Validation

The validation dataset is used to:

- Tune model parameters.
- Prevent overfitting.
- Select the best-performing model.

7. Model Evaluation

Evaluate the model using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

A high accuracy indicates reliable congestion prediction.

8. Deployment for Real-Time Decision Making

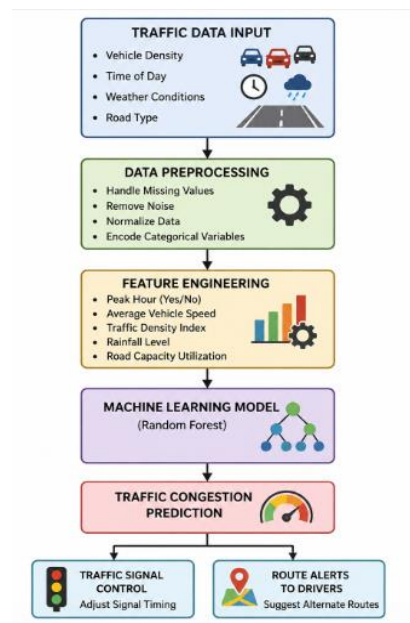
After successful evaluation, the model is deployed in the smart traffic management system.

The deployed model can:

- Predict traffic congestion in real time.
- Adjust traffic signal timings automatically.
- Suggest alternate routes to drivers.
- Send congestion alerts to traffic authorities.
- Support emergency vehicle routing.

Applications

- Smart traffic signal control.
- Real-time congestion prediction.
- Route optimization.
- Emergency response management.
- Reduced travel time and fuel consumption.



7. In the context of **Convolutional Neural Networks (CNNs)**, explain the different

layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

1. Convolution Layer

- Applies **filters (kernels)** to the input image.
- Detects features such as **edges, textures, and shapes**.
- Produces **feature maps** that highlight important regions of the image.

2. Pooling Layer

- Reduces the size of the feature maps.
- Removes unnecessary information.
- Decreases computation and helps prevent overfitting.

Example: Max Pooling selects the maximum value from each region.

3. Fully Connected Layer

- Flattens the extracted features into a vector.
- Combines all learned features.
- Produces the final classification result.

Example Output:

- Normal
- Pneumonia

4. Feature Map Generation

- The convolution layer slides a filter over the image.
- At each position, it computes feature values.
- The resulting output is called a **feature map**.

Example:

An X-ray image passes through edge-detection filters, generating feature maps that highlight lung boundaries and abnormal regions.

5. Backpropagation in CNN

Backpropagation updates the network weights during training.

Steps:

1. Perform a forward pass through the CNN.
2. Compare the predicted output with the actual label.
3. Calculate the prediction error.
4. Propagate the error backward.
5. Update filter weights using the learning rate.
6. Repeat until the error is minimized.

This process improves classification accuracy.

6. Example of Image Classification

- **Input:** Chest X-ray image.
- **Convolution Layer:** Detects edges and lung patterns.
- **Pooling Layer:** Reduces feature map size.
- **Fully Connected Layer:** Predicts whether the image is **Normal** or **Pneumonia**.

7. Factors Affecting Model Performance

- Quality and size of the training dataset.
- Number of convolutional layers.
- Filter size and number of filters.
- Learning rate.
- Activation function (ReLU).
- Pooling method.
- Batch size and number of training epochs.
- Regularization techniques (Dropout) to reduce overfitting.

8. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

1. Chromosome Representation

Each feature is represented as a binary gene.

Example:

Feature	Age	BP	Cholesterol	BMI	Smoking	Exercise
Chromosome 1	1	0		1	0	1

- **1** = Feature Selected
- **0** = Feature Not Selected

2. Fitness Evaluation

Each chromosome is evaluated by:

- Classification Accuracy
- Number of Selected Features

A chromosome with **high accuracy and fewer features** receives a higher fitness value.

3. Selection

- Selects chromosomes with the highest fitness.
- Better feature subsets are chosen as parents.

4. Crossover

Combines two parent chromosomes.

Example:

Parent 1 → **110101**

Parent 2 → **101111**

Child → **110111**

The offspring inherits useful features from both parents.

5. Mutation

Randomly changes a gene.

Example:

Before Mutation

110111

After Mutation

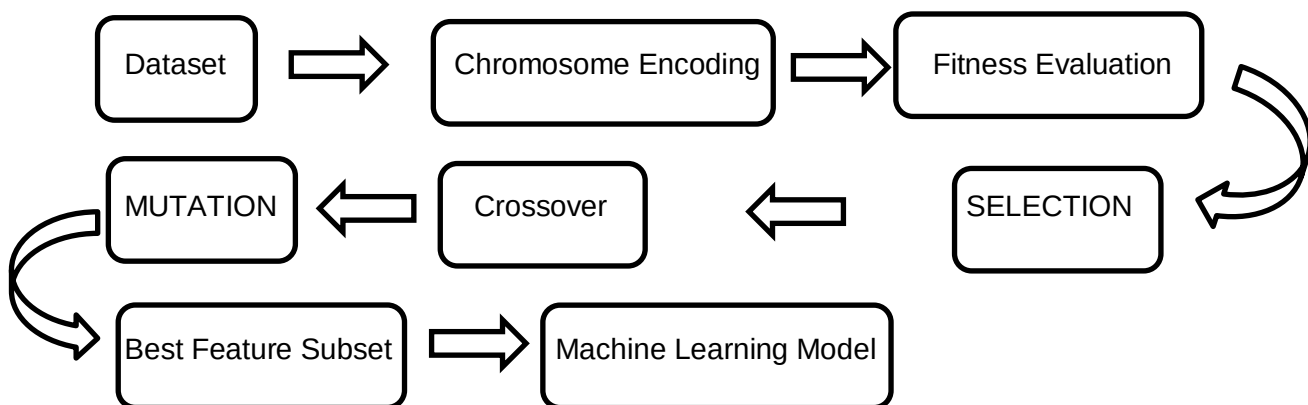
111111

Mutation introduces diversity and avoids local optima.

Benefits

- Removes irrelevant features.
- Improves prediction accuracy.
- Reduces training time.
- Prevents overfitting.
- Produces a simpler model.

SKETCH:



9. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

Scenario

A bank develops a machine learning model to detect **fraudulent transactions**. The model's performance is evaluated using different classification metrics.

Sample Confusion Matrix

	Predicted Fraud	Predicted Not Fraud
Actual Fraud	TP = 40	FN = 10
Actual Not Fraud	FP = 5	TN = 45

Where:

- **TP (True Positive):** Fraud correctly identified.
- **TN (True Negative):** Normal transaction correctly identified.
- **FP (False Positive):** Normal transaction predicted as fraud.
- **FN (False Negative):** Fraud predicted as normal.

Evaluation Metrics

1. Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$= (40 + 45) / 100 = 85\%$$

It measures the overall correctness of the model.

2. Precision

$$\text{Precision} = TP / (TP + FP)$$

$$= 40 / (40 + 5) = 88.9\%$$

It measures how many predicted fraud cases are actually fraud.

3. Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 40 / (40 + 10) = 80\%$$

It measures how many actual fraud cases are correctly detected.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall.

$$\text{F1-Score} = 84.2\%$$

It provides a balanced measure when both Precision and Recall are important.

5. ROC Curve

The ROC curve compares the **True Positive Rate** and **False Positive Rate**. A model with a larger **Area Under the Curve (AUC)** has better classification performance.

Applications

- **Medical Diagnosis:** High Recall helps detect most patients with the disease.
- **Fraud Detection:** High Precision reduces false fraud alerts.

10. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently.

Scenario

A company wants to schedule tasks among multiple processors to reduce execution time and improve processor utilization. A **Genetic Algorithm (GA)** is used to find the optimal task schedule.

1. Chromosome Encoding

Each chromosome represents one possible task schedule.

Example:

Tasks: T1, T2, T3, T4

Chromosome: [P1, P2, P1, P3]

where P1, P2, and P3 represent processors assigned to each task.

2. Fitness Evaluation

The fitness function evaluates each schedule based on:

- Total execution time
- Processor utilization
- Waiting time

Schedules with lower execution time and better resource utilization receive higher fitness values.

3. Selection

The best schedules are selected as parents according to their fitness values. Better schedules have a higher chance of producing the next generation.

4. Crossover

Two parent chromosomes exchange some of their task assignments to produce new offspring with improved schedules.

5. Mutation

Mutation randomly changes the processor assigned to a task. This introduces diversity into the population and helps avoid local optimum solutions.

Applications

- CPU Scheduling
- Cloud Computing
- Manufacturing Systems
- Project Scheduling
- Job Scheduling

Conclusion

Genetic Algorithms optimize task scheduling by representing schedules as chromosomes and improving them through **selection, crossover, and mutation**. The algorithm produces an optimal schedule that minimizes execution time, improves processor utilization, and increases overall system performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

